

Examination Seating Management System (ESMS)

Abstract

The Examination Seating Allocation System is a web-based application designed to automate the process of assigning students to examination halls and bench positions. Educational institutions often perform this task manually, which is time-consuming, error-prone, and difficult to manage, especially when multiple departments, subjects, examination sessions, and classrooms are involved.

The system allows administrators to import student information, classroom details, and examination schedules extracted from PDF documents. Students are automatically grouped by class and examination subject, arranged in ascending roll-number order, and assigned to available classrooms and bench positions according to institutional rules — including room capacity, bench seating limits (3 students per bench), and a maximum of three departments per room. The system also supports multiple examination sessions per day and accommodates classrooms with varying seating capacities.

At its core, the project is a **constraint-based multi-objective optimization problem**. The allocation engine enforces mandatory constraints such as seating capacity, bench occupancy, roll-number ordering, and unique seat assignments, while simultaneously optimizing competing objectives including room utilization, department distribution, minimization of department fragmentation, and reduction of rooms containing four or five departments. The engine is implemented as a framework-independent Python module, allowing it to be developed, tested, and validated independently of the Django web application and the PDF processing pipeline.

Developed using Python and the Django web framework, the application provides a centralized platform for managing examination seating efficiently. It generates room-wise seating plans, student-wise allocation reports, department summaries, and other printable documents required by examination coordinators — reducing administrative effort, minimizing human error, and ensuring fair, systematic student placement.

Team & Roles

Member	Role	Responsibilities
Jesso Joy - 25MCA027	Data & Pipeline Lead	Database schema & Django models, PDF ingestion/parsing, input validation, Django admin, auth, deployment & environment setup
Sajeer F M - 25MCA049	Allocation Engine & Output Lead	Core allocation engine (grouping, hard/soft constraints, overflow handling), engine unit testing, report generation (all 5 report types), frontend templates & UI

Tech Stack

Layer	Technology
Backend	Python, Django
Frontend	HTML, CSS, JavaScript, React
Database	PostgreSQL
Input processing	PDF parsing (e.g. pdfplumber / PyPDF2)
Testing	pytest
Version control	Git / GitHub
Output formats	Dynamic HTML tables, printable reports; PDF/Excel export (planned future enhancement)